



GOSFORD HIGH SCHOOL

2007

PRELIMINARY HIGHER SCHOOL CERTIFICATE

ASSESSMENT TASK 4

Yearly Examination

MATHEMATICS – EXTENSION 1

General Instructions:

- Reading time – 5 minutes
- Working time – 2 Hours
- Write using black or blue pen.
- Board-approved calculators may be used
- All necessary working should be shown in every question.

Total marks: - 72

- Attempt Questions 1 - 6
- All Questions are of equal value

QUESTION 1 (12 Marks)

- (a) Solve $\frac{5}{x-3} < 7$ and graph the solution on a number line 3
- (b) Find the acute angle between the lines $3x + 2y - 7 = 0$ and $y = 4x - 5$ to the nearest degree 3
- (c) Find the coordinates of the point that divides the interval A(2, 6) and B(9, 3) externally in the ratio 5:2 3
- (d) (i) Show that $\frac{1}{a^2 + ab} + \frac{1}{b^2 + ab} = \frac{1}{ab}$ 2
- (ii) Hence express $\frac{1}{7}$ in the form $\frac{1}{a} + \frac{1}{b}$ where a and b are positive integers 1

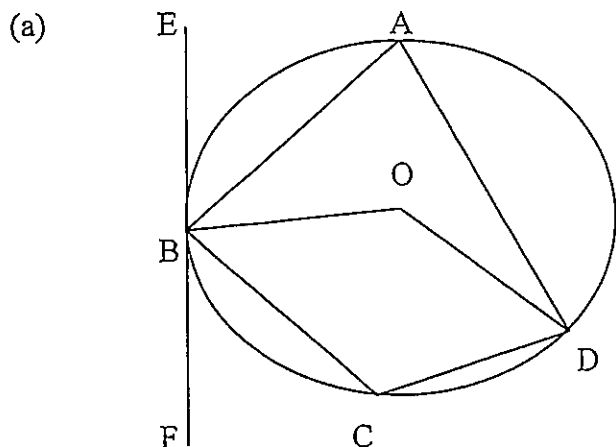
END OF QUESTION 1**QUESTION 2 (Start a New Page) (12 Marks)**

- (a) Calculate the remainder when $P(x) = x^3 + 2$ is divided by $x - 1$ 1
- (b) The polynomial $P(x) = x^3 + ax + 12$ has a factor $(x + 3)$. Find the value of a . 2
- (c) Find the quotient $Q(x)$ and the remainder, $R(x)$, when the polynomial $P(x) = x^4 - x^2 + 1$ is divided by $x^2 + 1$ 3
- (d) The point $P(3, 7)$ lies on the graph of the odd polynomial function $y = P(x)$. Find, with reasons,
- (i) the remainder when $P(x)$ is divided by $(x - 3)$ 1
- (ii) the remainder when $P(x)$ is divided by $(x + 3)$ 1
- (e) Solve $x^3 - 4x^2 - x + 4 = 0$ 2
- (f) If α , β and γ are roots of the cubic equation $5x^3 + 7x^2 - 3x - 4 = 0$, find the value of 2

$$\alpha^2 + \beta^2 + \gamma^2$$

END OF QUESTION 2

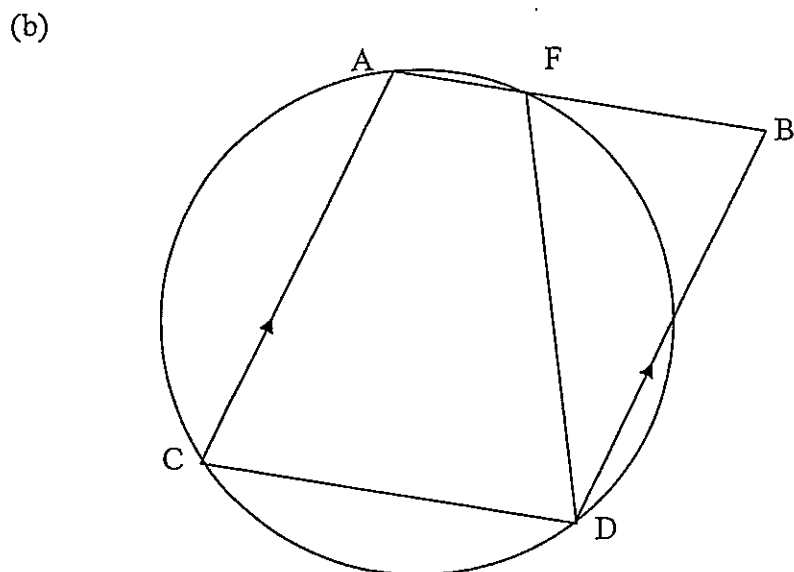
QUESTION 3 (Start a New Page) (12 Marks)



ABCD is a quadrilateral inscribed in a circle with centre O. Reflex $\angle BOD = 240^\circ$. Find, giving reasons

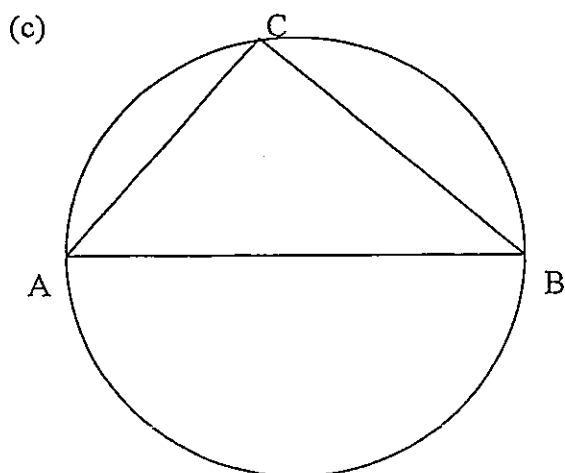
- (i) the size of $\angle BAD$
- (ii) the size of $\angle BCD$
- (iii) the size of $\angle FBD$

2
2
2



In the diagram (not to scale), $AC = BD$ and $AC \parallel BD$. Prove $DF = DB$

4



In the diagram (not to scale), $AB = 41$ cm, $AC = 9$ cm and $BC = 40$ cm
Prove AB is the diameter of the circle drawn

2

END OF QUESTION 3

QUESTION 4 (Start a New Page) (12 Marks)

- (a) By using an expression for $\tan(\alpha + \beta)$, find the exact value of $\tan 105^\circ$ 2
- (b) Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$ 3
- (c) Solve the equation $\sqrt{3} \sin \theta - \cos \theta = 1$ for $0 \leq \theta \leq 2\pi$ 4
- (d) A ship, S, is sailing on a straight course for an oil rig, R, on a bearing of 050° . At the same time, a lighthouse, L, is sighted in a direction of 012° . The charts indicate that the bearing from the rig to the lighthouse is 340° , and a distance of 18 km from it.
- (i) Draw a diagram showing the above information 1
- (ii) Find the distance from the ship to the oil rig correct to the nearest metre 2

END OF QUESTION 4**QUESTION 5 (Start a New Page) (12 Marks)**

- (a) Show $\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \sec \theta$ 2
- (b) Sketch $f(x) = ||x - 2| - 1|$ 2
- (c) Using First Principles, find the derivative of 2
- $f(x) = 4x^2 + 5$
- (d) For the curve $y = x^2(x^2 - 1)$,
- (i) Find the derivative $\frac{dy}{dx}$ 1
- (ii) Hence or otherwise, find the equation of the normal at the point (1, 0) 2
- (e) The straight line joining the centre O, of a circle to an external point T, cuts the circumference of the circle at Z. $TZ = 32$ cm and the tangent to the circle from T is 40 cm long.
- (i) Draw a diagram to represent this information 1
- (ii) Calculate the radius of the circle giving reasons for your answer 2

END OF QUESTION 5

QUESTION 6 (Start a New Page) (12 Marks)

(a) Solve for x : $|x^2 - 5| \leq 5x + 9$ 2

(b) The equation $x^3 + px^2 + qx + pq = 0$, where $p \neq 0$ and $q \neq 0$ has 3 real roots α , β and γ .

(i) By considering the relationships between the roots and the coefficients of the equation, show that $(\alpha + \beta + \gamma)\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right) = 1$ 2

(ii) Show that $-p$ is a root of the equation. Hence show that $q < 0$. 2

(c) The quadratic equation $x^2 - x + k = 0$, where k is a real number has 2 distinct, positive real roots.

(i) Show that $0 < k < \frac{1}{4}$ 2

(ii) Let the two roots be α and β . Show that $\alpha^2 + \beta^2 = 1 - 2k$ and deduce that $\frac{1}{2} < \alpha^2 + \beta^2 < 1$ 4

END OF EXAMINATION

QUESTION 1 (12 Marks)

(a) Solve $\frac{5}{x-3} < 7$ and graph the solution on a number line

Note: $x \neq 3$

For $x < 3$

$$5 > 7x - 21$$

$$26 > 7x$$

$$\frac{26}{7} > x$$

$$x < \frac{26}{7}$$

For $x > 3$

$$5 < 7x - 21$$

$$26 < 7x$$

$$\frac{26}{7} < x$$

$$x > \frac{26}{7}$$

Testing points

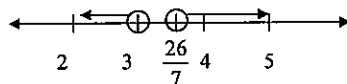
$$x \rightarrow 3^- \quad \frac{5}{x-3} < 7$$

$$x \rightarrow \frac{26^-}{7} \quad \frac{5}{x-3} > 7$$

$$x \rightarrow 3^+ \quad \frac{5}{x-3} > 7$$

$$x \rightarrow \frac{26^+}{7} \quad \frac{5}{x-3} < 7$$

$$\therefore x < 3 \text{ and } x > \frac{26}{7}$$



(b) Find the acute angle between the lines $3x + 2y - 7 = 0$ and $y = 4x - 5$ to the nearest degree

$$3x + 2y - 7 = 0$$

$$y = 4x - 5$$

$$\text{Gradient} = \frac{-3}{2}$$

$$\text{Gradient} = 4$$

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\tan \theta = \frac{-3 - 4}{1 + \frac{-3}{2} \times 4}$$

$$\tan \theta = \frac{-7}{1 - 6}$$

$$\tan \theta = \frac{7}{5}$$

$$\theta = 47.73^\circ$$

$$\theta = 48^\circ$$

(c) Find the coordinates of the point that divides the interval A(2, 6) and B(9, 3) externally in the ratio 5:2

The point (X, Y) which divides the interval in the ratio $r_1:r_2$ is given by

$$(X, Y) = \left(\frac{r_1 x_2 + r_2 x_1}{r_1 + r_2}, \frac{r_1 y_2 + r_2 y_1}{r_1 + r_2} \right)$$

$$(X, Y) = \left(\frac{5 \times 9 + (-2) \times 2}{5 + (-2)}, \frac{5 \times 3 + (-2) \times 6}{5 + (-2)} \right) \text{ Here } r_1 = 5 \text{ and } r_2 = -2 \text{ (externally)}$$

$$(X, Y) = \left(13\frac{2}{3}, 1 \right)$$

(d) (i) Show that $\frac{1}{a^2 + ab} + \frac{1}{b^2 + ab} = \frac{1}{ab}$

$$LHS = \frac{1}{a^2 + ab} + \frac{1}{b^2 + ab}$$

$$LHS = \frac{1}{a(a+b)} + \frac{1}{b(a+b)}$$

$$LHS = \frac{(a+b)}{ab(a+b)}$$

$$LHS = \frac{1}{ab}$$

$$\therefore LHS = RHS$$

(ii) Hence express $\frac{1}{7}$ in the form $\frac{1}{a} + \frac{1}{b}$ where a and b are positive integers

Choose factors of 7 $\rightarrow 1$ and 7 as a and b (from above) and write in reverse

$$\frac{1}{1 \times 7} = \frac{1}{1^2 + 1 \times 7} + \frac{1}{7^2 + 1 \times 7}$$

$$= \frac{1}{8} + \frac{1}{56}$$

END OF QUESTION 1

QUESTION 2 (Start a New Page) (12 Marks)

(a) Calculate the remainder when $P(x) = x^3 + 2$ is divided by $x - 1$

$$R(1) = 1^3 + 2$$

$$R(1) = 3$$

(b) The polynomial $P(x) = x^3 + ax + 12$ has a factor $(x + 3)$. Find the value of a .

2

$$R(-3) = (-3)^3 + a(-3) + 12$$

$$0 = -27 - 3a + 12$$

$$0 = -3a - 15 \quad \text{As } (x+3) \text{ is a factor, the remainder is 0 (i.e. } R(-3) = 0)$$

$$a = -5$$

(c) Find the quotient $Q(x)$ and the remainder, $R(x)$, when the polynomial $P(x) = x^4 - x^2 + 1$ is divided by $x^2 + 1$

3

$$\begin{array}{r} x^2 + 0x + 1 \overline{) x^4 + 0x^3 - x^2 + 0x + 1} \\ \underline{x^4 + 0x^3 + x^2} \\ -2x^2 + 0x + 1 \\ \underline{-2x^2 + 0x - 2} \\ 3 \end{array} \quad \therefore Q(x) = x^2 - 2$$

$$R(x) = 3$$

(d) The point $P(3, 7)$ lies on the graph of the odd polynomial function $y = P(x)$. Find, with reasons,

(i) the remainder when $P(x)$ is divided by $(x - 3)$

1

From the point $(3, 7)$ above, when $x = 3$, $y = 7$, therefore, $R(3) = 7$

(ii) the remainder when $P(x)$ is divided by $(x + 3)$

1

As the polynomial is odd, $P(x) = -P(-x)$, therefore, $R(-3) = -7$

2

(e) Solve $x^3 - 4x^2 - x + 4 = 0$

Firstly testing factors of 4 $\rightarrow$

$$P(1) = 1^3 - 4(1)^2 - 1 + 4 \rightarrow (x - 1) \text{ is a factor and } x = 1 \text{ is a solution}$$

$$P(1) = 0$$

$$P(-1) = (-1)^3 - 4(-1)^2 - (-1) + 4 \rightarrow (x + 1) \text{ is a factor } x = -1 \text{ is a solution}$$

$$P(-1) = 0$$

$$P(4) = 4^3 - 4(4)^2 - 4 + 4 \rightarrow (x - 4) \text{ is a factor } x = 4 \text{ is a solution}$$

$$P(4) = 0$$

$$\therefore (x + 1)(x - 1)(x - 4) = x^3 - 4x^2 - x + 4$$

OR

$$\therefore x = -1, 1, 4$$

(f) If α , β and γ are roots of the cubic equation $5x^3 + 7x^2 - 3x - 4 = 0$, find the value of

2

$$\alpha^2 + \beta^2 + \gamma^2$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - \alpha\beta - \alpha\gamma - \beta\alpha - \beta\gamma - \gamma\alpha - \gamma\beta$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right)$$

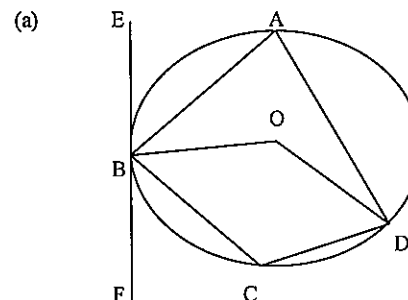
$$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{-7}{5}\right)^2 - 2\left(\frac{-3}{5}\right)$$

$$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{49}{25}\right) + \frac{6}{5}$$

$$\alpha^2 + \beta^2 + \gamma^2 = \frac{79}{25}$$

END OF QUESTION 2

QUESTION 3 (Start a New Page) (12 Marks)



ABCD is a quadrilateral inscribed in a circle with centre O. Reflex $\angle BOD = 240^\circ$. Find, giving reasons

(i) the size of $\angle BAD$

2

$$\angle BOD = 120^\circ (\angle \text{'s at a point given reflex } \angle BOD = 240^\circ)$$

$$\angle BAD = 60^\circ (\angle \text{ at centre} = \text{twice } \angle \text{ at circumference on same arc})$$

(ii) the size of $\angle BCD$

2

$$\angle BCD = 120^\circ (\text{given } \angle BOD = 240^\circ, \angle \text{ at centre} = \text{twice } \angle \text{ at circumference on same arc})$$

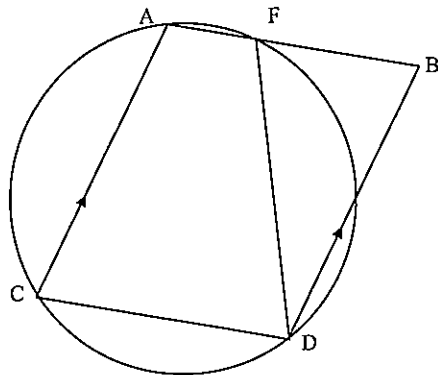
(iii) the size of $\angle FBD$

2

$$\angle FBD = \angle BAD (\angle \text{ between tangent and chord} = \angle \text{ in alternate segment})$$

$$\therefore \angle FBD = 60^\circ (\angle BAD = 60^\circ \text{ from above})$$

(b)



In the diagram (not to scale),

$AC = BD$ and $AC \parallel BD$. Prove $DF = DB$

4

$ABCD$ is a //ogram (given 1 pair opp sides = and //)

$\therefore \angle ACD = \angle ABD$ (opp $\angle$'s //ogram $ABCD =$)

$ACDF$ is a cyclic quad (quad with all vertices on the circumference of same circle)

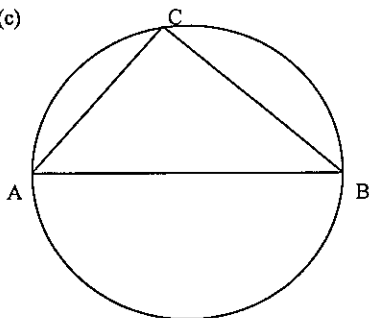
$\angle BFD = \angle ACD$ (ext $\angle$ cyc. quad. = $\angle$ in alt seg.)

$\therefore \angle BFD = \angle DBF$ (both = $\angle ACD$)

$\therefore \triangle DBF$ is isosc. (base $\angle$'s =)

$\therefore DF = DB$ (= sides opp = $\angle$'s of isosc $\triangle DBF$)

(c)



In the diagram (not to scale), $AB = 41$ cm,
 $AC = 9$ cm and $BC = 40$ cm
Prove AB is the diameter of the circle drawn

2

$$\begin{aligned} AB^2 &= 41^2 & AC^2 + BC^2 &= 40^2 + 9^2 \\ AB^2 &= 1681 & AC^2 + BC^2 &= 1681 \\ & & AC^2 + BC^2 &= AB^2 \end{aligned}$$

$\therefore \triangle ABC$ is a right angled triangle (square of hyp = sum of squares of other 2 sides)

$\therefore \angle ACB = 90^\circ$ (opp hypotenuse)

$\therefore AB$ is a diameter ($\angle$ at the circumference in a semi circle is 90°)

END OF QUESTION 3

QUESTION 4 (Start a New Page) (12 Marks)

(a) By using an expression for $\tan(\alpha + \beta)$, find the exact value of $\tan 105^\circ$

2

$$\tan 105^\circ = \tan(60^\circ + 45^\circ)$$

$$\tan 105^\circ = \frac{\tan 60^\circ + \tan 45^\circ}{1 - \tan 60^\circ \tan 45^\circ}$$

$$\tan 105^\circ = \frac{\sqrt{3} + 1}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$$

$$\tan 105^\circ = \frac{3 + 1 + 2\sqrt{3}}{1 - 3}$$

$$\tan 105^\circ = \frac{4 + 2\sqrt{3}}{-2}$$

$$\tan 105^\circ = -2 - \sqrt{3}$$

(b) Prove that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$

3

Using the t Method

$$\text{Let } \tan \theta = t \text{ then } \tan 2\theta = \frac{2t}{1 - t^2},$$

$$\sin 2\theta = \frac{2t}{1 + t^2} \text{ and } \cos 2\theta = \frac{1 - t^2}{1 + t^2}$$

Using Sum/Diff of Angles Method

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{\frac{2t}{1 + t^2}}{1 + \frac{1 - t^2}{1 + t^2}}$$

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{\frac{2t}{1 + t^2}}{\frac{1 + t^2 + 1 - t^2}{1 + t^2}}$$

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{\frac{2t}{1 + t^2}}{\frac{2}{1 + t^2}}$$

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{2t}{1 + t^2} \times \frac{1 + t^2}{2}$$

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = t$$

$$\therefore \frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$$

$$\begin{aligned} \frac{\sin 2\theta}{1 + \cos 2\theta} &= \frac{2 \sin \theta \cos \theta}{1 + 2 \cos^2 \theta - 1} \\ \frac{\sin 2\theta}{1 + \cos 2\theta} &= \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} \\ \frac{\sin 2\theta}{1 + \cos 2\theta} &= \frac{\sin \theta}{\cos \theta} \\ \therefore \frac{\sin 2\theta}{1 + \cos 2\theta} &= \tan \theta \end{aligned}$$

(c) Solve the equation $\sqrt{3} \sin \theta - \cos \theta = 1$ for $0 \leq \theta \leq 2\pi$

4

Using $a \sin \theta - b \cos \theta = r \sin(\theta - \alpha)$ where $r = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$, α acute

$$a = \sqrt{3}, b = -1 \rightarrow r = \sqrt{3+1} = 2 \quad \tan \alpha = \frac{-1}{\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

$$r = 2 \quad \alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \sin \theta - \cos \theta = 2 \sin(\theta - \frac{\pi}{6})$$

$$\therefore 2 \sin(\theta - \frac{\pi}{6}) = 1$$

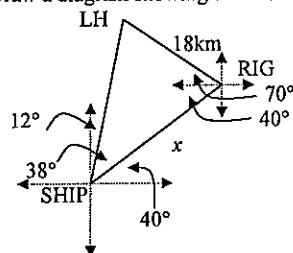
$$\sin(\theta - \frac{\pi}{6}) = \frac{1}{2}$$

$$\theta - \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{3}, 2\pi$$

(d) A ship, S, is sailing on a straight course for an oil rig, R, on a bearing of 050° . At the same time, a lighthouse, L, is sighted in a direction of 012° . The charts indicate that the bearing from the rig to the lighthouse is 340° , and a distance of 18 km from it.

(i) Draw a diagram showing the above information



(ii) Find the distance from the ship to the oil rig correct to the nearest metre

$$\frac{x}{\sin 32^\circ} = \frac{18}{\sin 38^\circ}$$

$$x = \frac{18 \sin 32^\circ}{\sin 38^\circ}$$

$$x = 15.493 \text{ km}$$

END OF QUESTION 4

QUESTION 5 (Start a New Page) (12 Marks)

(a) Show $\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \sec \theta$

2

$$\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \frac{\cos \theta \sin 2\theta}{\sin \theta \cos \theta} - \frac{\sin \theta \cos 2\theta}{\sin \theta \cos \theta}$$

$$\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \frac{\sin(2\theta - \theta)}{\sin \theta \cos \theta}$$

$$\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \frac{\sin \theta}{\sin \theta \cos \theta}$$

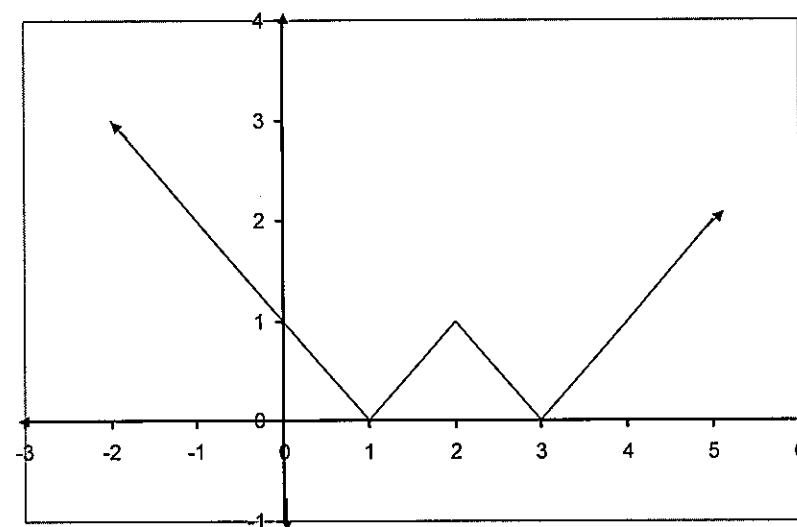
$$\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \frac{1}{\cos \theta}$$

$$\therefore \frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \sec \theta$$

(b) Sketch $f(x) = |x-2| - 1$

2

By inspection of critical points $x = 1$, $x = 2$, $x = 3$ and sketching



1

2

(c) Using First Principles, find the derivative of

$$f(x) = 4x^2 + 5$$

Let there be a small increase in x , by h

$$f(x+h) = 4(x+h)^2 + 5$$

$$f(x+h) = 4(x^2 + 2hx + h^2) + 5$$

$$f(x+h) = 4x^2 + 5 + 8hx + 4h^2$$

Now the gradient of the tangent is given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{4x^2 + 5 + 8hx + 4h^2 - (4x^2 + 5)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{8hx + 4h^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} 8x + 4h$$

$$f'(x) = 8x$$

(d) For the curve $y = x^2(x^2 - 1)$,

(i) Find the derivative $\frac{dy}{dx}$

$$\frac{dy}{dx} = 2x(x^2 - 1) + x^2(2x)$$

$$\frac{dy}{dx} = 2x(x^2 - 1 + x^2)$$

$$\frac{dy}{dx} = 2x(2x^2 - 1)$$

(ii) Hence or otherwise, find the equation of the normal at the point (1, 0)

At $x=1$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{-1}{2}(x - 1)$$

$$y = \frac{-x}{2} + \frac{1}{2}$$

or

$$x + 2y - 1 = 0$$

$$\frac{dy}{dx} = 2(2 - 1)$$

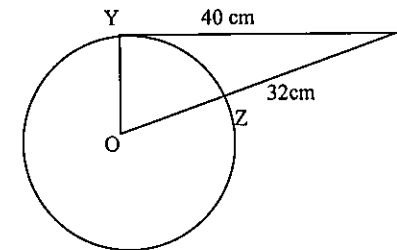
$$\frac{dy}{dx} = 2$$

$$\text{At } (1, 0) \text{ and } m = \frac{-1}{2}$$

2

(e) The straight line joining the centre O, of a circle to an external point T, cuts the circumference of the circle at Z. TZ = 32 cm and the tangent to the circle from T is 40 cm long.

(i) Draw a diagram to represent this information



(ii) Calculate the radius of the circle giving reasons for your answer

Let the radius of the circle be r and the pt of contact with the radius and the tangent be Y

$TY \perp OY$ (radius of circle $\perp$ tangent at pt of contact)

$$r^2 + 40^2 = (r + 32)^2$$

Therefore

$$r^2 + 1600 = r^2 + 64r + 1024$$

$$64r = 576$$

$$r = 9 \text{ cm}$$

END OF QUESTION 5

QUESTION 6 (Start a New Page) (12 Marks)

(a) Solve for x : $|x^2 - 5| \leq 5x + 9$

When $|x^2 - 5| > 0$

$$x^2 - 5 \leq 5x + 9$$

$$x^2 - 5x - 14 \leq 0$$

$$(x - 7)(x + 2) \leq 0$$

Critical Points

-2 and 7

Testing Points

$$\text{As } x \rightarrow -2^- \quad (x - 7)(x + 2) \geq 0$$

$$\text{As } x \rightarrow -2^+ \quad (x - 7)(x + 2) \leq 0$$

$$\text{As } x \rightarrow 7^- \quad (x - 7)(x + 2) \leq 0$$

$$\text{As } x \rightarrow 7^+ \quad (x - 7)(x + 2) \geq 0$$

$$\text{i.e. } -2 \leq x \leq 7$$

When $|x^2 - 5| < 0$

$$x^2 - 5 \geq -5x - 9$$

$$x^2 + 5x + 4 \geq 0$$

$$(x + 4)(x + 1) \geq 0$$

Critical Points

-4 and -1

$$\text{As } x \rightarrow -4^- \quad (x + 4)(x + 1) \geq 0$$

$$\text{As } x \rightarrow -4^+ \quad (x + 4)(x + 1) \leq 0$$

$$\text{As } x \rightarrow -1^- \quad (x + 4)(x + 1) \leq 0$$

$$\text{As } x \rightarrow -1^+ \quad (x + 4)(x + 1) \geq 0$$

$$\text{i.e. } -4 \geq x \geq -1$$

$$\therefore -1 \leq x \leq 7$$

1

2

1

2

2

(b) The equation $x^3 + px^2 + qx + pq = 0$, where $p \neq 0$ and $q \neq 0$ has 3 real roots α , β and γ .

(i) By considering the relationships between the roots and the coefficients of the equation, show that $(\alpha + \beta + \gamma)\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right) = 1$

2

$$(\alpha + \beta + \gamma) = \frac{-b}{a}$$

$$\alpha + \beta + \gamma = -p$$

$$\therefore (\alpha + \beta + \gamma)\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right) = -p \times \frac{-1}{p}$$

$$\therefore (\alpha + \beta + \gamma)\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right) = 1$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\frac{c}{-d}}{\frac{a}{-d}}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{q}{-pq}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{-1}{p}$$

(ii) Show that $-p$ is a root of the equation. Hence show that $q < 0$.

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If $-p$ is a root, then $P(-p) = 0$

$$P(-p) = (-p)^3 + p(-p)^2 + q(-p) + pq$$

$$P(-p) = -p^3 + p^3 - pq + pq$$

$$P(-p) = 0$$

Therefore $-p$ is a root

$$x^3 + px^2 + qx + pq = x^2(x + p) + q(x + p)$$

$$x^3 + px^2 + qx + pq = (x + p)(x^2 + q)$$

Therefore $P(x) = 0$ when $(x + p) = 0$ OR $(x^2 + q) = 0$

$$x^2 = -q$$

BUT

So either $x = -p$ OR $x^2 \geq 0$

$$\therefore -q > 0$$

$$\therefore q < 0$$

(c) The quadratic equation $x^2 - x + k = 0$, where k is a real number has 2 distinct, positive real roots.

(i) Show that $0 < k < \frac{1}{4}$

2

$$\Delta = b^2 - 4ac$$

$$\Delta = 1 - 4k$$

For 2 distinct, positive real roots, $\Delta > 0$ and the product of the roots is positive

$$1 - 4k > 0$$

$$1 > 4k$$

$$\text{So } \frac{1}{4} > k$$

$$k < \frac{1}{4}$$

$$\alpha\beta > 0$$

$$\text{and } \therefore \frac{c}{a} > 0$$

$$\therefore k > 0$$

$$\text{Therefore } 0 < k < \frac{1}{4}$$

(ii) Let the two roots be α and β . Show that $\alpha^2 + \beta^2 = 1 - 2k$ and deduce that

4

$$\frac{1}{2} < \alpha^2 + \beta^2 < 1$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$\alpha^2 + \beta^2 = 1^2 - 2k$$

$$\alpha + \beta = \frac{-b}{a}$$

$$\alpha + \beta = 1$$

From (i) above

$$0 < k < \frac{1}{4}$$

$$0 < 2k < \frac{1}{2}$$

$$0 > -2k > \frac{-1}{2}$$

$$1 > 1 - 2k > \frac{1}{2}$$

$$\frac{1}{2} < 1 - 2k < 1$$

$$\frac{1}{2} < \alpha^2 + \beta^2 < 1$$

END OF EXAMINATION